

ELEC 3EY4 - FINAL CHEAT SHEET

Amaam Ul Haque
400520641

Graduate Attributes

COURSE OBJECTIVES

By the end of this course, students should be able to:

- Understand and use basic commands in the Linux operating system
- Understand the principles of the Robot Operating System (ROS) and use it to develop and implement a real-time control system on Nvidia Jetson Nano AI embedded computer
- Analyze the magnetic and electrical characteristics of electric motor drives
- Use reference frame transformations to develop a model of a permanent magnet synchronous motor
- Understand the basic principles of sensing, planning, and control in self-driving vehicles
- Use coordinate frames and rigid transformations for expressing configuration and movement of rigid bodies in two and three-dimensional space
- Understand the principles of localization and mapping in mobile robotic systems
- Understand and develop feedback controllers for autonomous driving
- Understand principles of Quadratic Optimization and use it for developing an autonomous driving algorithm

Outcomes	Indicators	Measurement Method(s)
Actively contributes to the planning and execution of a team project.	6.1	Lab assignments and reports
Manages interpersonal relationships, taking leadership responsibilities as needed.	6.2	Lab assignments and reports
Demonstrates comprehension of technical and non-technical instructions and questions.	7.1	Lab assignments and reports
Composes an effective written document for the intended audience.	7.2	Lab reports
Composes and delivers an effective oral presentation for the intended audience.	7.3	Lab and exam questions
Describes the duty of a Professional Engineer to the public, client, employer, and the profession.	8.1	Exam questions
Evaluates the social impact of engineering activities, including health, safety, legal, cultural, and other relevant factors, and identifies uncertainties in decisions.	9.2	Lab assignments and reports
Plans and effectively manages a project's time, resources, and scope, following business practices as appropriate.	11.2	Lab assignments and reports
Identifies, characterizes, assesses, and manages risks to project success.	11.3	Lab assignments and reports
Reflects on one's educational needs and opportunities for growth.	12.1	Lab and exam questions
Seeks and acquires appropriate external information as required, including showing awareness of sources of information and ability to critically evaluate them.	12.2	Lab and exam questions

Vehicle Kinematics

$$\mathbf{X} = [x \ y \ \theta]^T \text{ w.r.t odom}$$

$$\dot{x} = v_s \cos \theta \rightarrow x_{i+1} = x_i + \Delta t \cdot \dot{x}_i$$

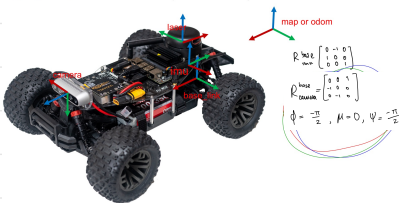
$$\dot{y} = v_s \sin \theta \rightarrow y_{i+1} = y_i + \Delta t \cdot \dot{y}_i$$

$$\dot{\theta} = \frac{v_s}{L} \tan \delta \rightarrow \theta_{i+1} = \theta_i + \Delta t \cdot \dot{\theta}_i$$

- IMU measures vehicle heading w.r.t earth attached frame & measures vehicle linear acceleration

Coordinate Frames

X, Y, and Z axes are denoted by corresponding colors



Solution:

Q1)

$$a) R_B^A = R_A \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} R_A \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{bmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$b) R_B^A = (R_B^B)^T \Rightarrow y_B^A = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 0 \end{bmatrix}$$

$$Q2) p^A = R_B^A p^B + O_B^A = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} + \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ -1 \end{bmatrix}$$

Q1. The frames A, B are initially aligned. Frame B undergoes two consecutive rotations, first about its own x by $\frac{\pi}{4}$ and then the z axis of Frame A by $\frac{\pi}{4}$. After these two rotations:

- Find the rotation matrix of Frame B with respect to Frame A, R_B^A .
- Find the y axis of Frame A, expressed in Frame B.

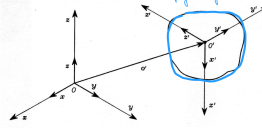
Q2. The Rotation and translation of Frame B with respect to Frame A are given by:

$$R_B^A = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, O_B^A = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

A point coordinates in Frame B is given by $p^B = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$. Find its coordinates in Frame A, $p^A = ?$.

Spatial Description of Rigid Bodies & Coord Frame

- Position and orientation of a rigid body is completely described in space by its position and orientation with respect to a right-handed reference frame.

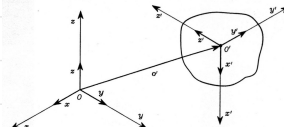


Position of a point O' on the body:

$$O' = o'_x x + o'_y y + o'_z z,$$

$$O' = \begin{bmatrix} o'_x \\ o'_y \\ o'_z \end{bmatrix}$$

- Orientation of the body: Attach an orthonormal frame $O'-x'y'z'$ to the body at O' . Find the orientation of the body frame $O'-x'y'z'$ wrt. the reference frame $O-xyz$.



$$x' = x'_x x + x'_y y + x'_z z$$

$$y' = y'_x x + y'_y y + y'_z z$$

$$z' = z'_x x + z'_y y + z'_z z$$

The parameters z'_x, z'_y, z'_z are the projection of z' along the x, y and z directions, that is

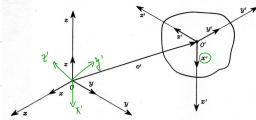
$$z'_x = \langle z', x \rangle, z'_y = \langle z', y \rangle, z'_z = \langle z', z \rangle$$

$$x'_x + x'_y + x'_z = 1$$

$$y'_x + y'_y + y'_z = 1$$

- Rotation Matrix R :

$$R = \begin{bmatrix} x' & y' & z' \end{bmatrix} = \begin{bmatrix} x'_x & y'_x & z'_x \\ x'_y & y'_y & z'_y \\ x'_z & y'_z & z'_z \end{bmatrix} = \begin{bmatrix} x'^T x & y'^T x & z'^T x \\ x'^T y & y'^T y & z'^T y \\ x'^T z & y'^T z & z'^T z \end{bmatrix}$$



Property: The column vectors of rotation matrix R are mutually orthogonal and have unit length:

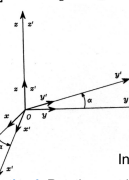
$$x'^T y' = 0 \quad y'^T z' = 0 \quad z'^T x' = 0,$$

$$x'^T x' = 1 \quad y'^T y' = 1 \quad z'^T z' = 1.$$

Therefore, $R^T R = I_{3 \times 3}$ or $R^{-1} = R^T \Rightarrow R$ is always an orthogonal matrix.

Elementary Rotations: Consider the body frame $O'-x'y'z'$ is obtained by rotating the reference frame $O-xyz$ by angle α wrt. the z axis (Figure). Then:

$$x' = \begin{bmatrix} \cos \alpha \\ \sin \alpha \\ 0 \end{bmatrix} \quad y' = \begin{bmatrix} -\sin \alpha \\ \cos \alpha \\ 0 \end{bmatrix} \quad z' = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \Rightarrow R_z(\alpha) = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Similarly, for elementary rotations by angles γ and β about axes x and y :

$$R_x(\gamma) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \gamma & -\sin \gamma \\ 0 & \sin \gamma & \cos \gamma \end{bmatrix}$$

$$R_y(\beta) = \begin{bmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{bmatrix}$$

In general, $R_k(\gamma) = R_k^{-1}(\gamma) = R_k^T(\gamma)$, $k = x, y, z$

Interpretation: Rotation matrix R describes the rotation about an axis in space needed to align the reference frame $O-xyz$ with the body frame $O'-x'y'z'$.

Example 1: $O'-x'y'z'$ is $O-xyz$ rotated by angle α about z axis (Figure 1):

$$p = R p' \quad R_z(\alpha) = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad p' = \begin{bmatrix} p'_x \\ p'_y \\ p'_z \end{bmatrix}$$

$$p_x = p'_x \cos \alpha - p'_y \sin \alpha$$

$$p_y = p'_x \sin \alpha + p'_y \cos \alpha$$

$$p_z = p'_z$$

$$p = p'_x$$

Example 2 (Rotation of Vectors): Consider vector p in the xy plane that is obtained by rotating a vector p' about the z axis of the reference frame $O-xyz$ by angle α (Fig. 2). In this case, from Figure 2

$$p_x = |p| \cos(\beta) \quad p_y = |p| \sin(\beta) \quad p_z = |p| \cos(\alpha + \beta) \quad p'_x = p'_x \cos \alpha - p'_y \sin \alpha \quad p'_y = p'_x \sin \alpha + p'_y \cos \alpha \quad p'_z = p'_z$$

$$p = R_z(\alpha) p'$$

Example: Figure below shows a frame "B" which is rotated relative to the frame "A" about z axis by $\alpha = 30$ degrees, and translated 10 units in X_A and 5 units in Y_A directions. Find p_A given $p_B = [3.0, 7.0, 0.0]^T$.

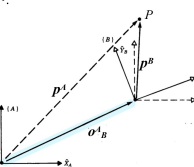
$$R_z(\alpha) = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad O_B^A = [10 \ 5 \ 0]^T$$

$$A_B^A = \begin{bmatrix} 0.866 & -0.500 & 0.000 & 10.0 \\ 0.500 & 0.866 & 0.000 & 5.0 \\ 0.000 & -0.000 & 1.000 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$p^B = [3.0, 7.0, 0.0, 1]^T$$

$$p^A = A_B^A p^B \Rightarrow p^A = [9.1, 12.6, 0.0, 1]^T$$

$$p^A = [9.1, 12.6, 0.0]^T$$



$$p^A = O_B^A + R_B^A p^B = A_B^A p^B$$

Mapping & Localisation

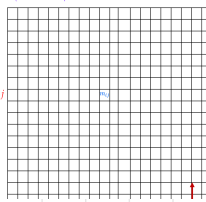
- Wheel odometry yields the vehicle pose $x_k = (\theta_k, d_{b_k}^0)$ at t_k , i.e., the transformation between the frames **odom** and **base**:

$$T_{b_k}^{odom} = \begin{bmatrix} R_{b_k}^0(\theta_k) & d_{b_k}^0 \\ 0 & 1 \end{bmatrix}$$

$\hat{x}_k = \begin{bmatrix} \cos \theta_k & -\sin \theta_k \\ \sin \theta_k & \cos \theta_k \end{bmatrix}$ ← angle / orientation
 $d_{b_k}^0 = \begin{bmatrix} x_k \\ y_k \end{bmatrix}$ ← magnitude / distance
 o horizontally polar coordinates of $\hat{x}_k$

Occupancy Grid Mapping

- The cell value $0 \leq p(m_{ij}) \leq 1$ is probability of cell m_{ij} being occupied:
 $p(m_{ij}) \geq p_{occ} > 0.5$ cell is occupied
 $p_{free} < p(m_{ij}) < p_{occ}$ cell is unknown
 $p(m_{ij}) \leq p_{free} < 0.5$ cell is free
free threshold
- Probability of cell being free:
 $p(\bar{m}_{ij}) = 1 - p(m_{ij})$



Inverse Sensor Model

- LIDAR scan returns q_k^i as coordinates of obstacle points in environment w.r.t. the **lidar** frame
 - These points are projected onto the fixed **odom** frame using the vehicle current pose $T_{b_k}^0$.
- $T_{b_k}^0 = \begin{bmatrix} \cos \theta_k & -\sin \theta_k & 0 \\ \sin \theta_k & \cos \theta_k & 0 \\ 0 & 0 & 1 \end{bmatrix}$
 $q_k^i = \begin{bmatrix} x_k^i \\ y_k^i \\ 1 \end{bmatrix}$ ← lidar point
 $\hat{q}_k^i = T_{b_k}^0 T_{b_k}^b \begin{bmatrix} q_k^i \\ 1 \end{bmatrix}$ ← odom point
 $\hat{q}_k^i = \begin{bmatrix} x_k^i \cos \theta_k - y_k^i \sin \theta_k \\ x_k^i \sin \theta_k + y_k^i \cos \theta_k \\ 1 \end{bmatrix}$
- For each cell m_{ij} , we find the closed LIDAR ray to the line connecting the cell to the origin of LIDAR frame
 - Cell is considered **occupied** if LIDAR return from this ray falls within cell
 - Cell is considered **free** if LIDAR return from this ray has a longer range than the cell distance to origin of LIDAR frame
 - Cell occupancy status is presumed unknown otherwise

Cell Occupancy Update

- We are interested in the finding cell occupancy probabilities given **all** LIDAR observations $z_{0:k}$ and vehicle poses $x_{0:k}$, i.e.,
 $p(m_{ij}|z_{0:k}, x_{0:k}) = ?$ ← prob. of cell occupancy based on all measurements of lidar from 0 to k, vehicle positions
- Using the **Bayes' rule**,
 $p(m_{ij}|z_{0:k}, x_{0:k}) = \frac{p(z_k|m_{ij}, x_k)p(m_{ij}|z_{0:k-1}, x_{0:k-1})}{p(z_k|z_{0:k-1}, x_{0:k-1})}$
- Using a **Markov Model assumption** → only x_k , no past data
 $p(z_k|m_{ij}, x_k) = p(z_k|x_k)$
 $p(m_{ij}|z_{0:k}, x_{0:k}) = \frac{p(z_k|x_k)p(m_{ij}|z_{0:k-1}, x_{0:k-1})}{p(z_k|z_{0:k-1}, x_{0:k-1})}$

Cell Occupancy Update

- By dividing the the two conditional probabilities:

$$\frac{p(m_{ij}|z_{0:k}, x_{0:k})}{1 - p(m_{ij}|z_{0:k}, x_{0:k})} = \frac{p(m_{ij}|z_k, x_k)}{1 - p(m_{ij}|z_k, x_k)} \cdot \frac{p(m_{ij}|z_{0:k-1}, x_{0:k-1})}{1 - p(m_{ij}|z_{0:k-1}, x_{0:k-1})}$$

if cell uncertain before about map → prior knowledge
- Let,

$$l(y) = \log \frac{p(y)}{1 - p(y)}$$
- With

$$p(y) = 1 - \frac{1}{1 + e^{l(y)}}$$

Cell Occupancy Update

- We obtain the following **recursive** update rule for the map occupancy probabilities:

$$l(m_{ij}|z_{0:k}, x_{0:k}) = l(m_{ij}|z_k, x_k) + l(m_{ij}|z_{0:k-1}, x_{0:k-1}) - l(m_{ij})$$

inverse sensor model recursive term prior

- Usually,

$$l(m_{ij}) = \log \frac{p_{un}}{1 - p_{un}} = \log \frac{0.5}{1 - 0.5} = 0$$

$p = 1$: $\log = \infty$ → cell is **occupied**
 $p = 0$: $\log = -\infty$ → cell is **free**
 $p = 0.5$: $\log = 0$ → cell is **unknown**

Finding The Transformation

- In summary, the vehicle configuration at t_{k+1} is given by:
 $T_{b_{k+1}}^0 = T_{b_k}^0 T_{b_k}^{b_{k+1}}$
- $T_{b_k}^0$ is the vehicle configuration in space at time t_k and is known
- The change in vehicle configuration in space from t_k to t_{k+1} , $T_{b_k}^{b_{k+1}}$, is unknown. It can be estimated from **Scan Matching** of the points

$$T_{b_{k+1}}^{b_k} = T_{b_k}^{b_{k+1}} T_{b_k}^0 = ?$$

- We would like to estimate $T_{b_{k+1}}^{b_k}$
- Ideally, for each pair of **corresponding** points from t_k and t_{k+1} ,

$$p_x^k = T_{b_{k+1}}^{b_k} p_x^{k+1}$$

- This can also be written as:

$$\hat{p}_x^k = R_{b_{k+1}}^{b_k} \hat{p}_x^{k+1} + d_{b_{k+1}}^k$$

$$T_{b_{k+1}}^{b_k} = \begin{bmatrix} R_{b_{k+1}}^{b_k} & d_{b_{k+1}}^k \\ 0 & 1 \end{bmatrix}$$

Simultaneous Localization and Mapping (SLAM)

- This a nonlinear optimization that can be solved iteratively

$$\arg \min_{\theta} \sum_j [1 - M(q_j^i)]^2$$

$$\begin{bmatrix} q_j^i \\ 1 \end{bmatrix} = T_{b_k}^0(\hat{x}_k) T_{b_k}^b \begin{bmatrix} q_j^i \\ 1 \end{bmatrix}$$

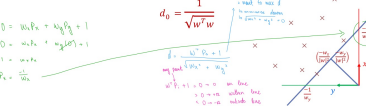
$$\hat{x}_k = (\hat{\theta}_k, \hat{d}_{b_k}^0) = x_{k-1} + \delta x$$

$$T_{b_k}^0 = \begin{bmatrix} \cos \hat{\theta}_k & -\sin \hat{\theta}_k & \hat{d}_{b_k}^0 \\ \sin \hat{\theta}_k & \cos \hat{\theta}_k & \hat{d}_{b_k}^0 \\ 0 & 0 & 1 \end{bmatrix}$$

Optimisation

Separating Virtual Barrier

- We want to find the line with **maximum distance** to the origin of LIDAR frame that separates obstacle points from it
- The line equation is given by:
 $w^T p + 1 = 0$, $w = \begin{bmatrix} w_x \\ w_y \end{bmatrix} \in \mathbb{R}^2$, $p = \begin{bmatrix} p_x \\ p_y \end{bmatrix} \in \mathbb{R}^2$
- The distance of origin to the line is:
 $d_0 = \frac{1}{\sqrt{w_x^2 + w_y^2}}$



- Note that for a point p_i behind the virtual barrier line:

$$w^T p_i + 1 \leq 0$$

- The solution to the following Inequality Constrained Quadratic Program (IQP) yields the desired virtual barrier for a set of obstacle points $\{p_i\}$:

$$\min_w \frac{1}{2} w^T w$$

$$\text{s.t. } w^T p_i + 1 \leq 0 \quad \forall i$$

w → each obstacle point / lidar scan

- Line equation for the right barrier:

$$w^T p + b = 1$$

- Line equation for the left barrier:

$$w^T p + b = -1$$

- Gap between left and right barriers:

$$-1 < b < 1$$

→ vehicle range between $(-1, 1)$

$$d = \frac{2}{\sqrt{w^T w}}$$

- Optimization problem to find **parallel** virtual barrier:

$$\min_{w, b} \frac{1}{2} w^T w$$

$$\text{s.t. } w^T p_i + b - 1 \geq 0 \quad \forall i$$

$$w^T p_j + b + 1 \leq 0 \quad \forall j$$

$$-1 + \epsilon \leq b \leq 1 - \epsilon$$

- A Quadratic Program has the following general form:

$$\min_x \frac{1}{2} x^T P x + q^T x + r$$

$$\text{s.t. } Gx \leq h$$

$$Ax = b$$

- $P \in \mathbb{R}^{n \times n}$, $P > 0$ (P is a positive definite matrix)
- $G \in \mathbb{R}^{m \times n}$, $A \in \mathbb{R}^{p \times n}$

Equality Constrained QP (EQP)

- We consider a Quadratic Optimization with **equality** constraints:

$$\min_x \frac{1}{2} x^T P x + q^T x + r$$

$$\text{s.t. } Ax = b$$

- The **Lagrangian** function for this optimization problem is defined as

$$L(x, v) = \frac{1}{2} x^T P x + q^T x + r + v^T (Ax - b)$$

- where $v \in \mathbb{R}^p$ is a vector of so-called **Lagrange multipliers**

EQP

- The **KKT conditions** for optimality are given by:
 $\nabla L(x^*, v^*) = 0$

- This yields

$$Ax^* = b$$

$$Px^* + q + A^T v^* = 0$$

- Or

$$\begin{bmatrix} P & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} x^* \\ v^* \end{bmatrix} = \begin{bmatrix} -q \\ b \end{bmatrix}$$

- This is a set of $n + p$ linear equations involving $n + p$ unknowns $\begin{bmatrix} x^* \\ v^* \end{bmatrix}$; it is the **KKT system of equations** for Quadratic Optimization with equality constraints

- Example:**

$$\min_x f(x) = 5x_1^2 + 8x_2^2 + 4x_1x_2 - 2x_1 + 4x_2 + 10$$

$$= \frac{1}{2} \begin{bmatrix} x_1 & x_2 \end{bmatrix}^T \begin{bmatrix} 10 & 4 \\ 4 & 16 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + [-2 \quad 4] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + 10$$

$$\text{s.t. } x_1 + 2x_2 = 1$$

$$P = \begin{bmatrix} 10 & 4 \\ 4 & 16 \end{bmatrix}, \quad q = [-2 \quad 4]$$

$$A = \begin{bmatrix} 1 & 2 \end{bmatrix}, \quad b = 1$$

$$n = 2, p = 1$$

- Solution to KKT equations:

$$\begin{bmatrix} P & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} x^* \\ v^* \end{bmatrix} = \begin{bmatrix} -q \\ b \end{bmatrix}$$

$$\begin{bmatrix} 10 & 4 & 1 \\ 4 & 16 & 2 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1^* \\ x_2^* \\ v^* \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x_1^* \\ x_2^* \\ v^* \end{bmatrix} = \begin{bmatrix} 10 & 4 & 1 \\ 4 & 16 & 2 \\ 1 & 2 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 2 \\ -4 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.6 \\ 0.2 \\ -4.8 \end{bmatrix}$$

$$f(x_1^*, x_2^*) = 12.2$$

Consider the following Quadratic Optimization with an equality constraint:

$$\min_{x_1, x_2} x_1^2 + 4x_2^2 - 3x_1 + x_2 + 3$$

$$\text{s.t. } 2x_1 - 3x_2 = 4$$

Use the Lagrangian function to derive a set of linear equations the solution to which yields the optimal point x_1^* , x_2^* . (Do not need to solve the equations)

Solution:

$$L(x, v) = x_1^2 + 4x_2^2 - 3x_1 + x_2 + 3 + v(2x_1 - 3x_2 - 4)$$

$$\nabla L(x^*, v^*) = 0$$

$$2x_1^* - 3x_2^* = 4$$

$$2x_1^* + 2v^* = 3$$

$$8x_2^* - 3v^* = -1$$